

2014 NCKU Top-Notch Interim Report

Research on resources circulation of critical metals

Subtitle: Applications of using glass fiber to controlled low strength material in civil engineering

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Outline

1.Content of Research

2.Experiment Design

3.Test Results

4.Conclusion



1. Content of Research

- ◆ The current reuse of waste circuit card is focus on the mental part but not the substrates.
- ◆ The substrates are consisted of glass fibers. Although many studies of using glass fibers to civil engineering have been done, the substrates are consisted of **very short fibers**. This study aims to discuss the possibility of using very short fiber to the CLSM.
- ◆ Main items of this study are shown as follows:



1.1 Preliminary application planning

◆ Application of glass fiber as an construction material

- (a) Establish the best ratio of fiber reinforced concrete.
- (b) Feasibility assessment in fiber reinforced concrete.

◆ Product development in fiber reinforced concrete

- (a) High strength of concrete products: such as interlocking bricks etc.
- (b) Low strength of concrete products: such as light compartment plates, controlled low strength material (CLSM) etc.



1.2 Feasibility study in controlled low strength material

◆ Physical and chemistry properties of glass fiber

- (a) Investigate the chemistry properties of glass fiber.
- (b) Investigate the physical properties of glass fiber.

◆ To establish the best mixed proportion of glass fiber

- (a) CLSM design strength: 30kgf/cm², 50kgf/cm²
- (b) Slump flow test: 45cm.
- (c) Fiber content: 10%, 20%, 30%, 50%.
- (d) Find the best mixed proportion in CLSM.



2. Experiment Design

◆ XRF test

◆ SEM test

◆ Compressive strength test

1. ASTM C109
2. Design strength: 30kgf/cm², 50kgf/cm²
3. Fiber content: 10%, 20%, 30%, 50%.

◆ Bending test

1. ASTM C348
2. Design strength: 30kgf/cm², 50kgf/cm²
3. Fiber content: 10%, 20%, 30%, 50%.



3. Test results

◆ XRF of glass fiber

Main compounds	Content
Al	> 611100 ± 7500 ppm
Si	329000 ± 1800 ppm
Ca	28520 ± 80 ppm
Cu	> 10620 ± 20 ppm
Br	> 16620 ± 20 ppm
As	< 4.0 ppm
Cd	< 30 ppm
Cr	< 26 ppm
Hg	< 7.9 ppm
Ni	< 7.5 ppm
Pb	< 8.6 ppm
Zn	< 6.4 ppm

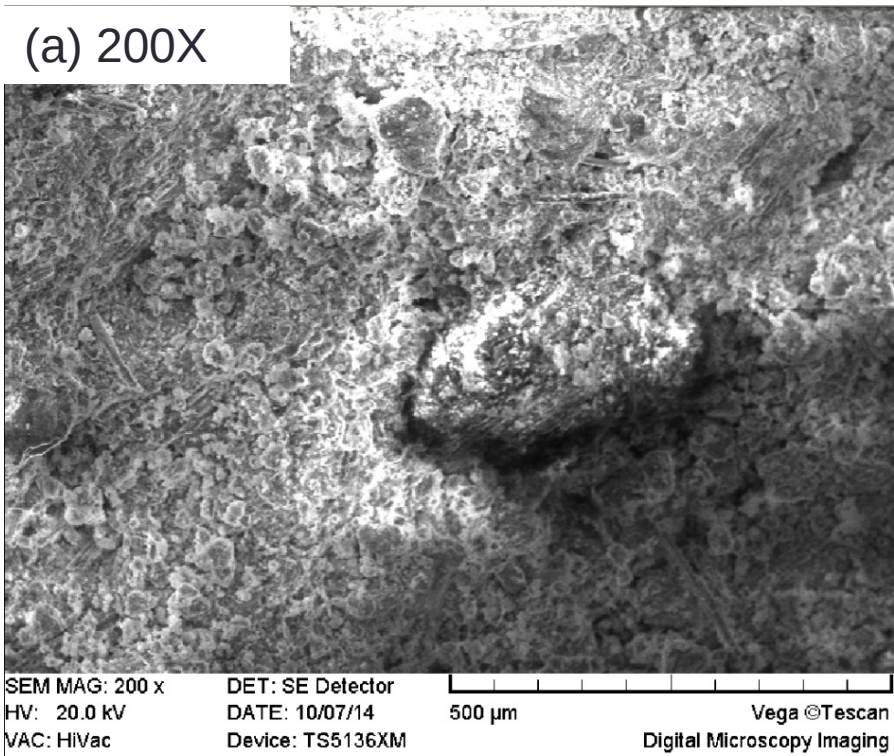
Element	Control criteria of heavy metal	
	Farm	Food farm
As	60 mg/kg	---
Cd	20 mg/kg	5 mg/kg
Cr	250 mg/kg	---
Cu	400 mg/kg	200mg/kg
Hg	20 mg/kg	5mg/kg
Ni	200 mg/kg	---
Pb	2000 mg/kg	500mg/kg
Zn	2000 mg/kg	600mg/kg

1.EPA, Taiwan(2001)

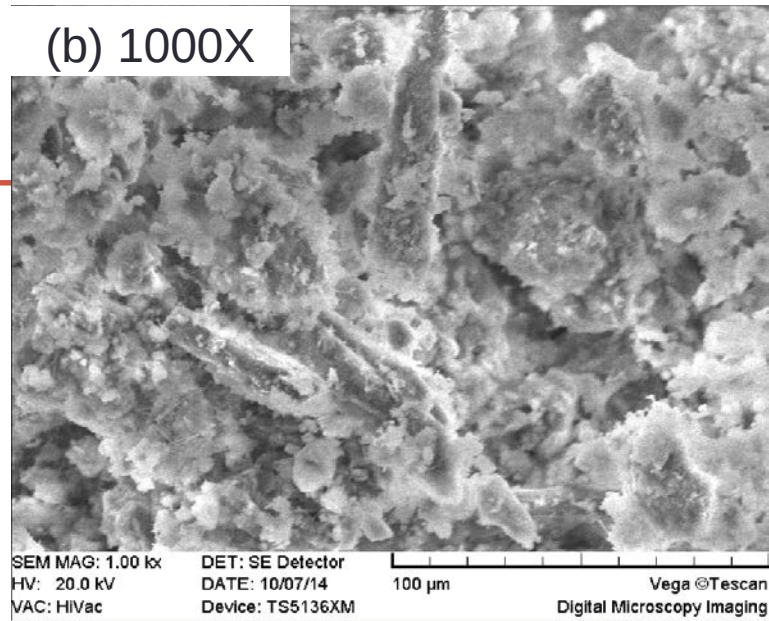
◆ SEM test

◆ The sample of CLSM (C30F10)

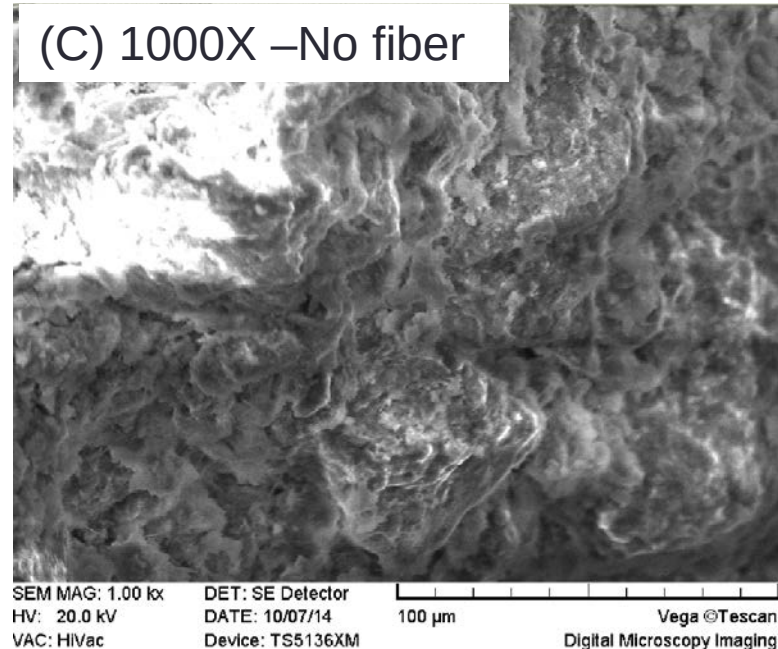
(a) 200X



(b) 1000X



(c) 1000X –No fiber

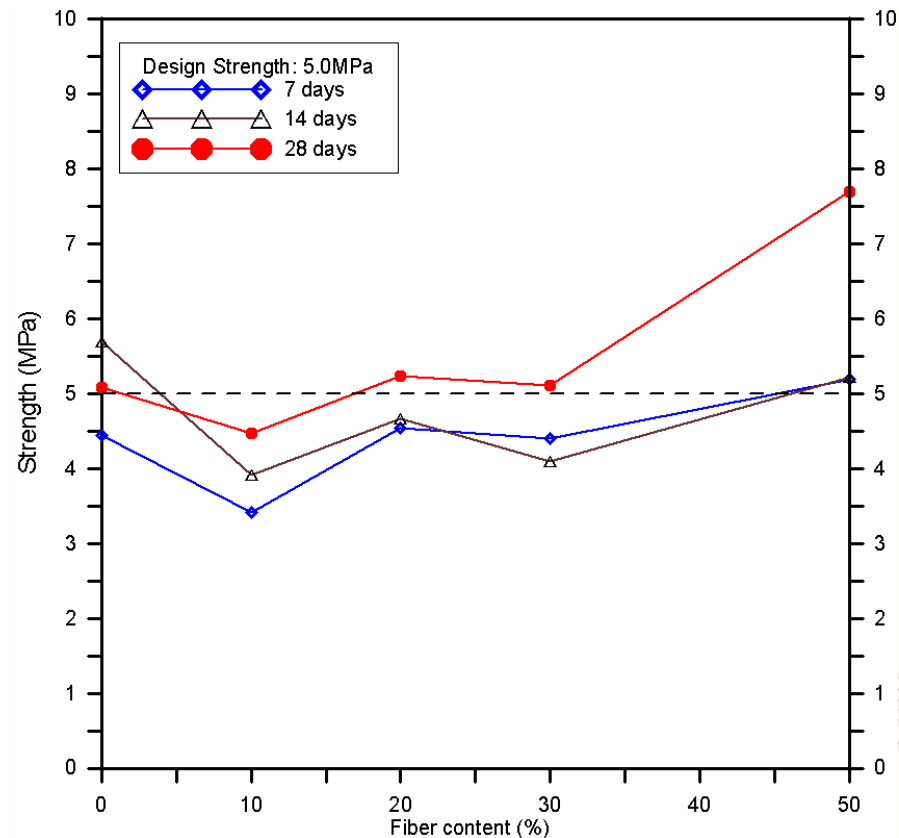
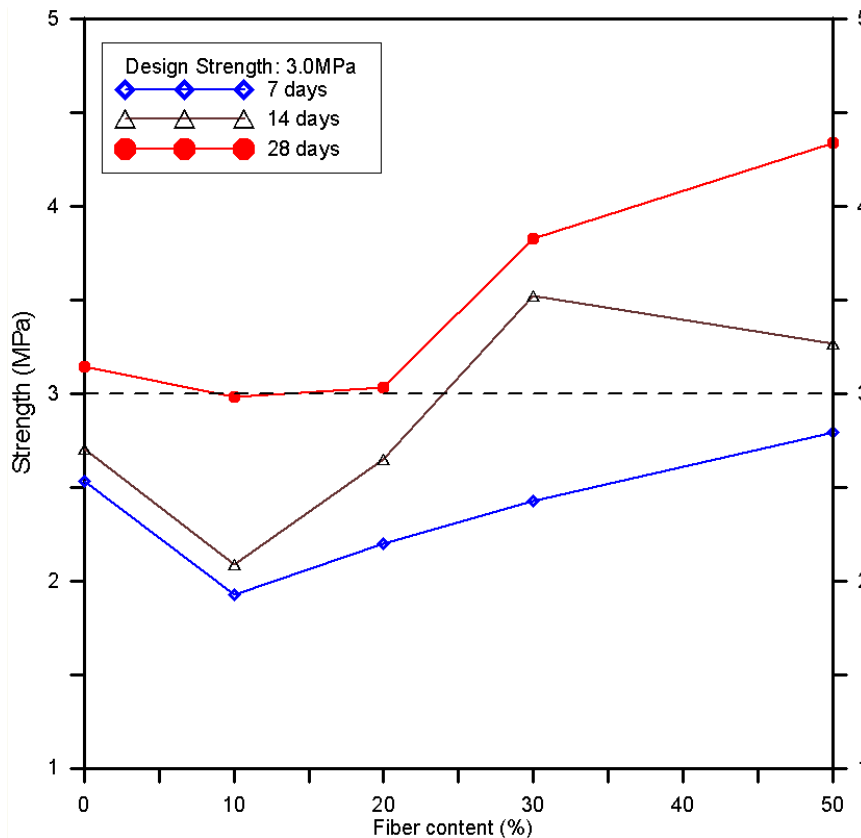


◆ The compressive strength of CLSM

Water-cement ratio (W/C)	Fiber content (%)	Unit weight (t/m ³)	Compressive strength (MPa)			Design strength (MPa)
			7 days	14 days	28 days	
2.0	0	1.89	2.535	2.705	3.145	3.0 (Ages: 28 days)
	10	1.87	1.925	2.090	2.980	
	20	1.81	2.200	2.650	3.035	
	30	1.84	2.425	3.520	3.830	
	50	1.69	2.795	3.265	4.340	
1.5	0	1.99	4.445	5.695	5.085	5.0 (Ages: 28 days)
	10	1.87	3.410	3.910	4.475	
	20	1.91	4.545	4.670	5.235	
	30	1.80	4.395	4.100	5.105	
	50	1.69	5.195	5.215	7.695	

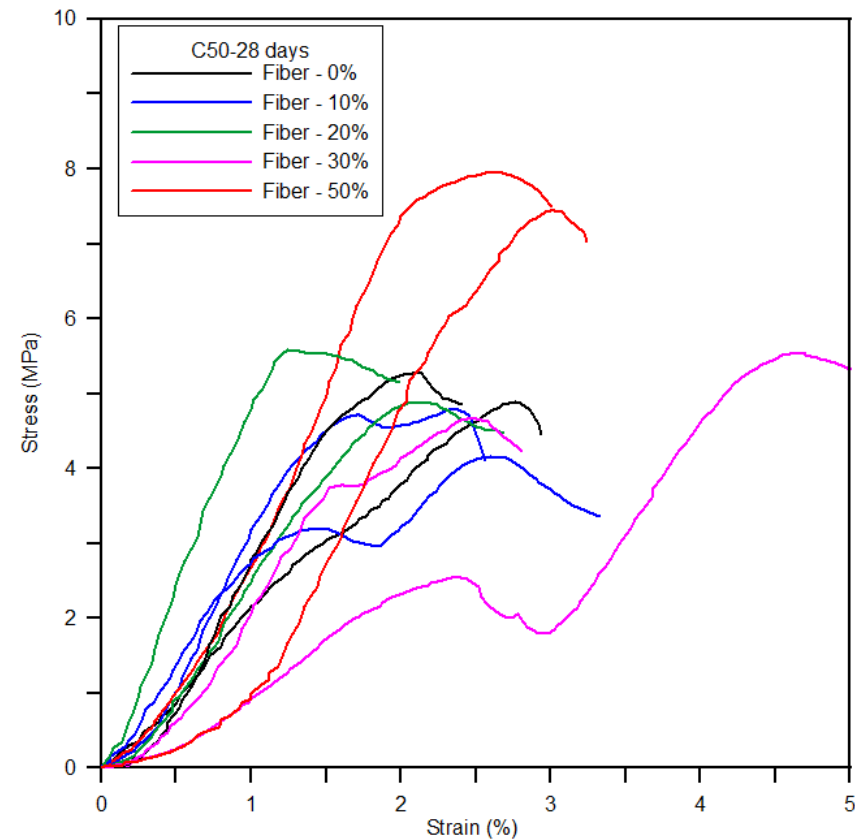
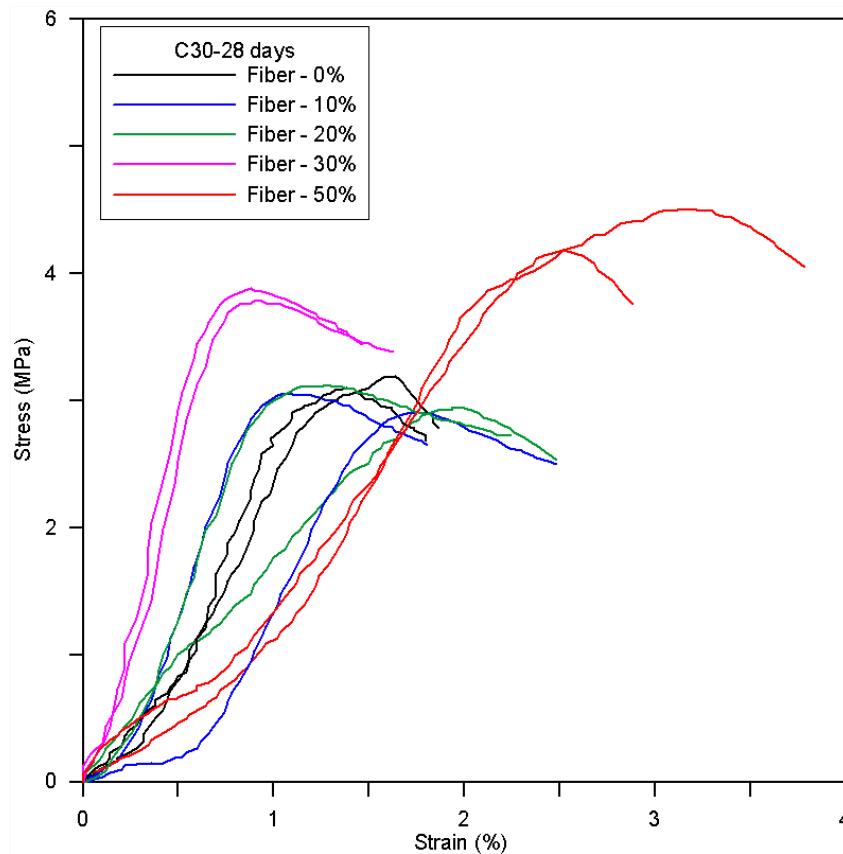
✂ The compressive strength of CLSM

1. The 28-day compressive strength of 10% fiber of the CLSM can not meet the requirements.
2. The compressive strength of 50% fiber of the CLSM at 28 days are better than others.



✂ The stress-strain curve of CLSM

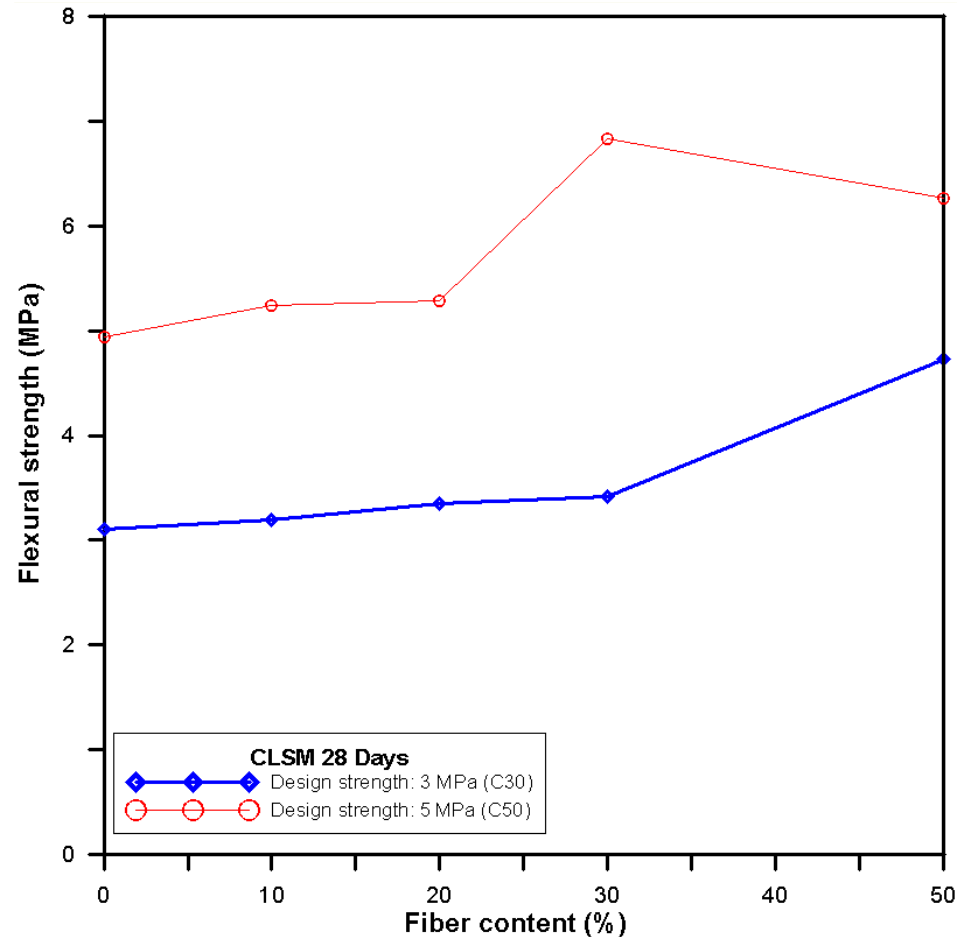
- ◆ Glass fiber content is higher, the ductility behavior of CLSM is more obvious.



❖ The flexural strength of CLSM

◆ Glass fiber content is higher, the flexural strength of CLSM is more higher.

Water-cement ratio (W/C)	Fiber content (%)	Flexural strength (MPa)
2.0	0	3.11
	10	3.20
	20	3.35
	30	3.41
	50	4.73
1.5	0	4.94
	10	5.24
	20	5.28
	30	6.84
	50	6.27



4. Conclusion

- ◆ Glass fibers from substrates are very short and have the limitation on applications. Therefore, they are used as fine aggregates.
- ◆ The content of copper can not meet the control criteria of heavy metal.
- ◆ The compressive strength of 50% fiber of the CLSM at 28 days are better result than others. The higher glass fiber content, the strength of CLSM is also higher.
- ◆ For 28-day flexural strength of the CLSM, the higher glass fiber content, the strength of CLSM is also higher.
- ◆ The glass fiber can be used in CLSM, but need to notice the high copper content.





The End



Thanks for your attention